



SYSTEMS THINKING: DECONSTRUCTING A UPS

An analysis of parts, interactions, and feedback loops within an Uninterruptible Power Supply.

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THE WHOLE SYSTEM

WHAT IS A UPS?

An Uninterruptible Power Supply (UPS) is an electrical apparatus that provides emergency power to a load when the input power source or mains power fails.

- **Primary Function:** Protective buffer.
- **Goal:** Prevent data loss and hardware damage.
- **Input:** Raw Grid Power.
- **Output:** Stabilized Backup Power.



Fig 1.0: The UPS Unit before dismantling

THE "BLACK BOX" PERSPECTIVE



INPUTS

AC Mains (230V/110V)
User Commands (Power
Button)



PROCESS

Rectification (AC to DC)
Energy Storage
(Charging)
Inversion (DC to AC)



OUTPUTS

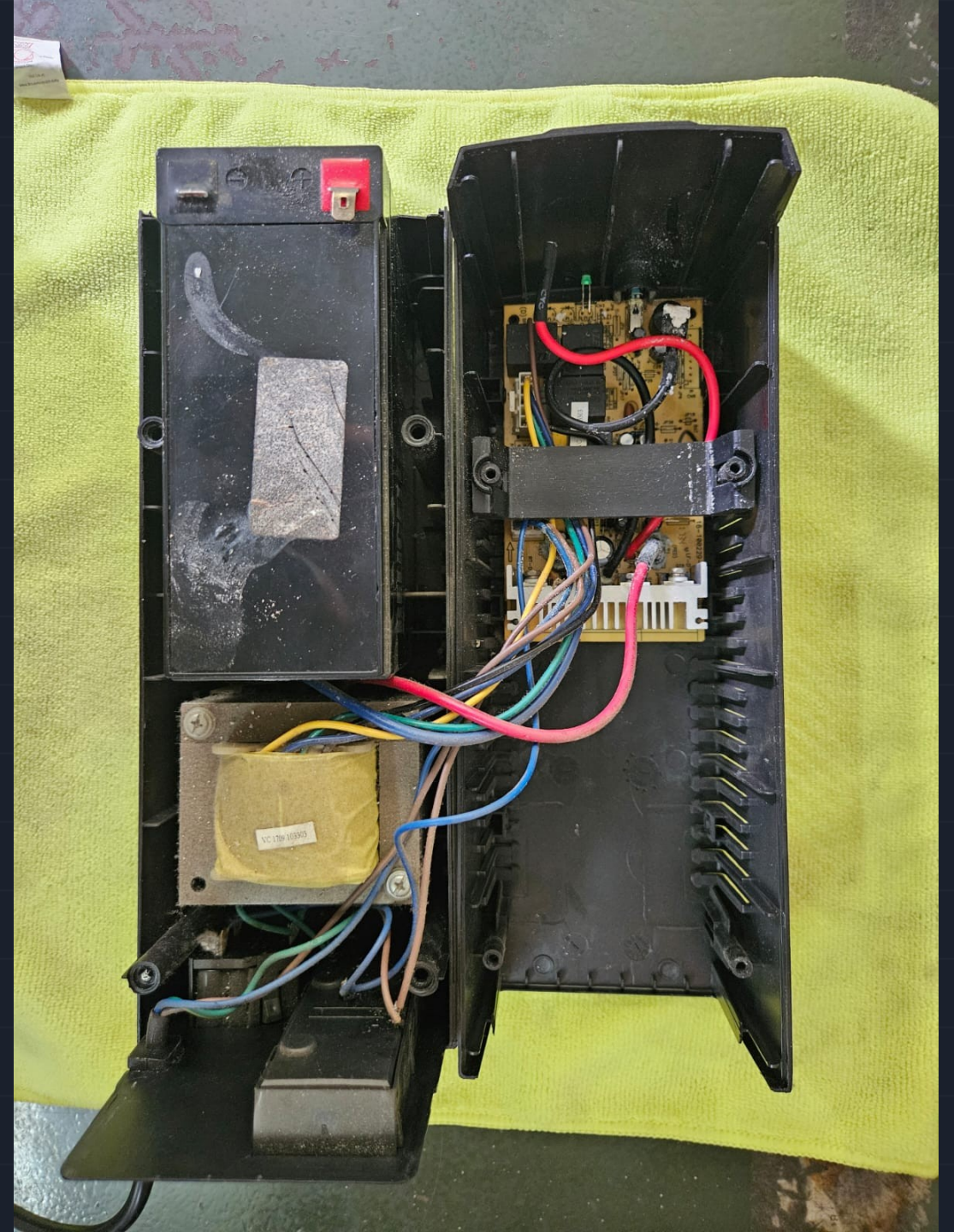
Clean AC Power
Status Indicators
(LED/Beep)

BREAKING THE BOUNDARY

Dismantling the external casing is the first step in systems analysis. It removes the abstraction layer intended for the user, revealing the internal complexity.

Inside, we see how the components are packed for thermal management and safety.

Task: Remove 4 chassis screws to reveal internal architecture.



SUBSYSTEM 1: ENERGY STORAGE



Fig 2.0: 12V 7Ah Sealed Lead Acid Battery

THE HEART OF THE SYSTEM

The battery is the chemical reservoir of potential energy.

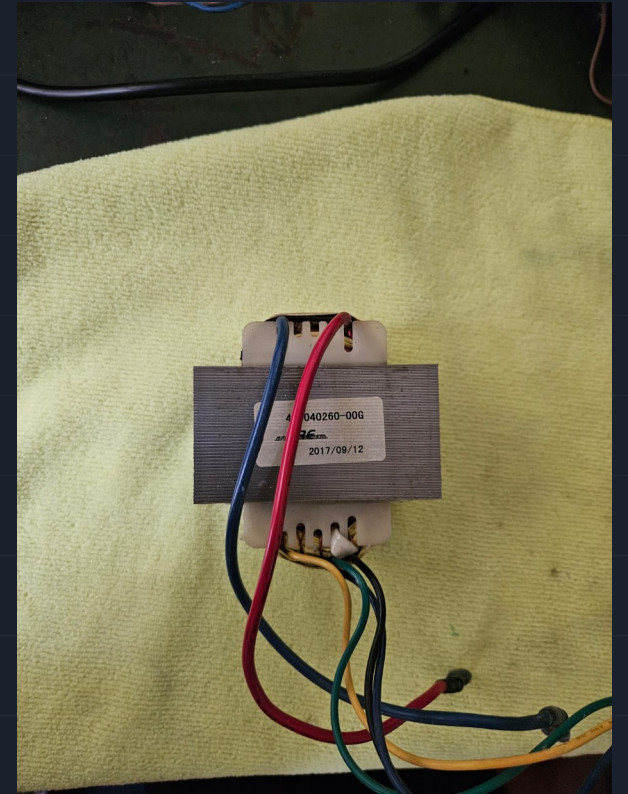
- **Type:** Sealed Lead Acid (SLA).
- **Role:** Accumulator. It stores energy during normal operation (charging) and releases it during failure (discharging).
- **System Dynamic:** Acts as the "stock" in the stock-and-flow diagram of energy.

SUBSYSTEM 2: THE TRANSFORMER

VOLTAGE REGULATION

The heaviest component in the system. It uses electromagnetic induction to change voltage levels.

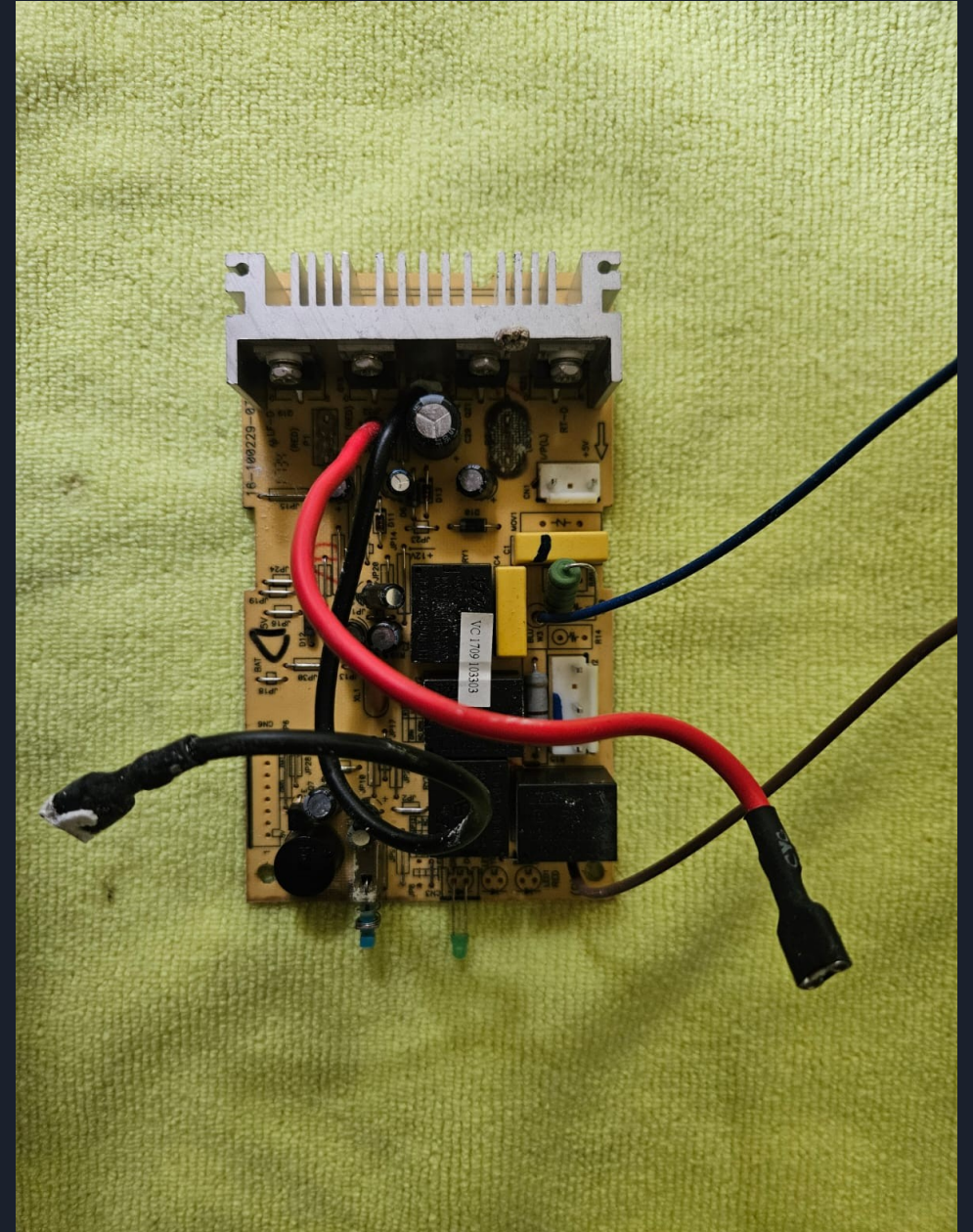
- **Step-Down:** Lowers mains voltage to charge the 12V battery safely.
- **Step-Up:** Raises the 12V battery output back to 230V/110V for your devices during an outage.
- **Material:** Copper coils around an iron core.



THE BRAIN: CIRCUIT BOARD (PCB)

This printed circuit board contains the logic and switching mechanisms.

- **Inverter Circuit:** Oscillates DC current to simulate AC waves.
- **Relays:** Physical switches (black boxes) that click to change power paths.
- **Microcontroller:** The decision maker. It monitors voltage inputs and triggers the battery mode within milliseconds.



INTERCONNECTIONS & FLOW



THE NERVOUS SYSTEM

Components alone do not make a system; their connections do.

- **Thick Red/Black Wires:** Handle high current from the battery.
- **Fuses:** Safety valves. If current exceeds limits, the fuse blows to protect the expensive transformer and battery.
- **Feedback Loop:** Sensors on the PCB constantly read the output voltage to adjust the inverter frequency.

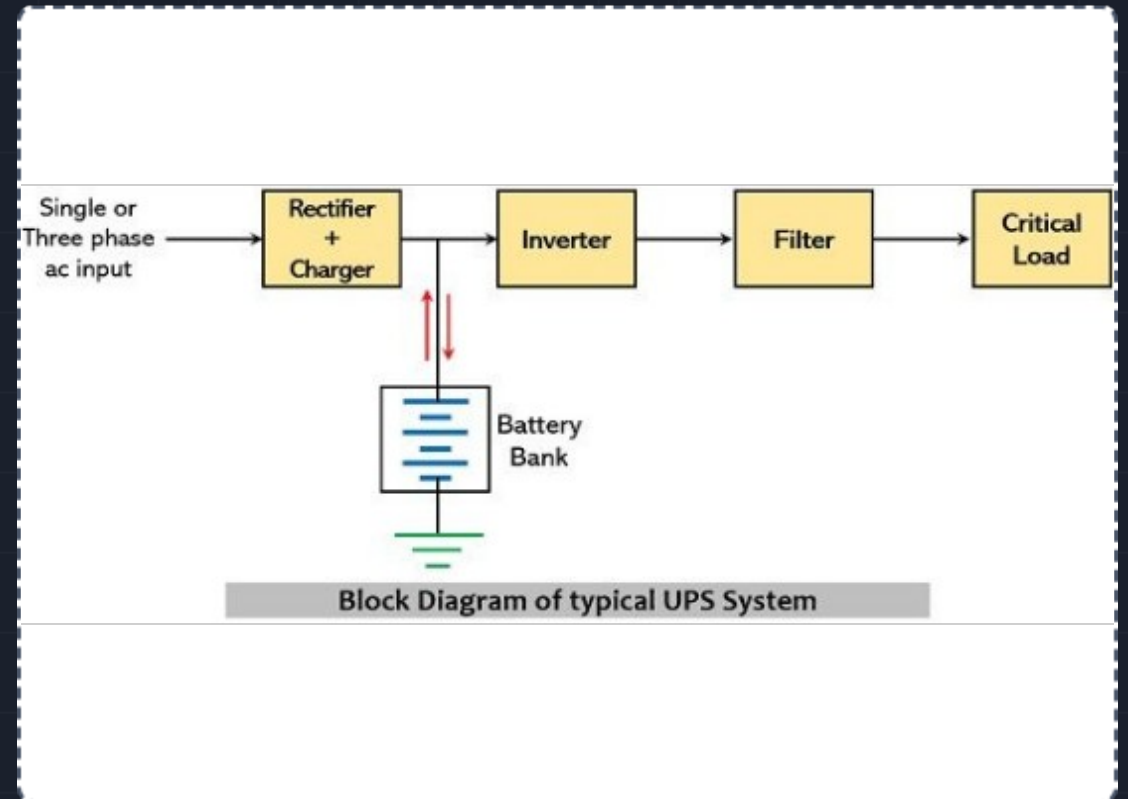
VISUALIZING THE LOGIC

BLOCK DIAGRAM

This diagram represents the logical flow of energy and information.

Note how the "Transfer Switch" is the critical decision point, determined by the "Control Logic" observing the "AC Input".

This visualizes the invisible rules governing the system.



SYSTEMS THINKING INSIGHTS



REDUNDANCY

The system is designed with a backup energy source that sits idle 99% of the time, highlighting the cost of reliability.



FEEDBACK

The "clicking" sound of a UPS is a mechanical feedback loop in action—relays physically moving to reroute power.



EMERGENCE

Neither a battery nor a transformer alone can provide "uninterrupted power." The capability emerges only from their interaction.

THANK YOU